

On-Chip Active High- Q Slow Light Topological Cavities

Xin Zhao, Yi Ji Tan, Wenhao Wang, Kaiji Chen, Zhonglei Shen, Shixiong Wang, Jianjia Yi,*
Lina Zhu,* and Ranjan Singh*

Topological photonic waveguides for lossless transport through edge states have enabled the design of cavities with sharp bends in a closed-loop configuration, where the confinement of photons in topological cavities has typically been controlled by varying the cavity size. Here, a slow light topological cavity is presented, where the light confinement and the quality (Q) factor are controlled through engineered dispersion of the slow light waveguide mode and its group velocity, introducing a new degree of freedom to design high- Q photonic cavities. It is experimentally demonstrated that topological slow light in valley photonic crystals with bearded interfaces enhances light-matter interactions, resulting in improved cavity Q -factor. In addition, active Q -factor control of slow light topological cavities is achieved through optical excitation of the silicon cavity chip, offering promising applications in communications, sensing, and polaritonics.

coherence integrated lasers, paving the way for applications such as coherent communications, quantum communications computing, and position and rotation detection.^[2,8] The measured Q -factor is proportional to the lifetime of the electromagnetic waves trapped within the resonators, which implies that a higher Q -factor leads to a longer interaction time between light and matter, and hence the enhanced light-matter interactions.^[3] Consequently, slow light promotes stronger light-matter interactions^[9] and offers a way to increase the Q -factor.^[4] In addition, a slower wave requires a shorter device length to achieve the same level of functionalities such as a phase delay, which encourages the exploitation of slow light for reducing device footprints.^[10]

1. Introduction

High quality factor (Q -factor) electromagnetic resonators, particularly on-chip devices, have been extensively studied for their potential to enhance nonlinear effects^[1,2] and light-matter interactions,^[3,4] for applications including sensing,^[5,6] communication,^[7] and lasing.^[2,8] The performance of these devices for on-chip sensing and frequency isolation can be significantly improved by increasing their Q -factors. For instance, sensors with higher Q -factors allow high-sensitivity detection of the complex permittivity of analytes due to the enhanced electromagnetic fields of the resonant mode.^[6] Resonators with higher Q -factors also have the potential to realize unprecedented high-

The discovery of photonic topological insulators (PTIs) has attracted attention for their topological edge states and corner states,^[11–15] where the edge states are robust against back reflections even at sharp bends, corners, and in the presence of unit cell defects. Recent works have reported topological cavities^[3,4,16–18] and slow light modes^[10,19–28] in PTI systems for realizing applications in nonlinear topological photonics,^[15] topological quantum circuits,^[29] terahertz communications,^[30–35] and ultra-low threshold topological lasers.^[4,36] Several studies have achieved broadband slow light in topological waveguides fabricated from magneto-optical photonic crystals by selectively exciting slow

X. Zhao, L. Zhu
State Key Lab of Integrated Services Networks
Xidian University
Xi'an 710071, China
E-mail: lnzhu@xidian.edu.cn

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X. Zhao, Y. J. Tan, W. Wang, K. Chen, Z. Shen
Division of Physics and Applied Physics
School of Physical and Mathematical Sciences
Nanyang Technological University
21 Nanyang Link, Singapore, Singapore, 637371

X. Zhao, Y. J. Tan, W. Wang, K. Chen, Z. Shen
Centre for Disruptive Photonic Technologies
The Photonics Institute
Nanyang Technological University
21 Nanyang Link Singapore, Singapore, 637371

S. Wang, J. Yi
School of Electronic and Information Engineering
Xi'an Jiaotong University
Xi'an 710049 China
E-mail: jianjia.yi@xjtu.edu.cn

R. Singh
Department of Electrical Engineering
University of Notre Dame
Notre Dame, IN 46556, USA
E-mail: rsingh3@nd.edu

Table 1. Comparison of this study with existing topological cavities and slow light works.

With topological cavity	High-Q	Slow light	On-chip	Require magnetic field	Active control	Low loss	Frequency	Experiment
No ^[3]	Yes	No	No	No	No	No	optical	No
Yes ^[4]	Yes	Yes	No	No	No	Yes	optical	Yes
Yes ^[16]	Yes	No	Yes	No	No	No	optical	Yes
Yes ^[17]	Yes	No	Yes	No	Yes	Yes	terahertz	Yes
Yes ^[18]	Yes	No	Yes	No	No	Yes	terahertz	Yes
No ^[19]	No	Yes	No	Yes	Yes	No	microwave	No
No ^[20]	No	Yes	No	Yes	Yes	No	microwave	No
No ^[21]	No	Yes	No	No	No	No	–	No
No ^[22]	Yes	Yes	No	No	No	No	optical	Yes
No ^[23]	No	Yes	No	No	No	No	optical	No
No ^[24]	No	Yes	No	No	No	No	optical	Yes
No ^[25]	No	Yes	No	No	No	No	–	No
No ^[26]	No	Yes	No	No	No	No	–	No
No ^[27]	No	No	Yes	No	No	Yes	terahertz	No
No ^[28]	No	Yes	Yes	No	Yes	Yes	terahertz	Yes
Yes ^[32]	Yes	No	Yes	No	No	No	terahertz	Yes
Yes ^[33]	Yes	No	Yes	No	Yes	No	terahertz	Yes
Yes ^[34]	No	No	Yes	No	No	No	terahertz	Yes
Yes ^[35]	Yes	No	Yes	No	Yes	No	terahertz	Yes
No ^[37]	No	Yes	No	Yes	No	No	–	No
No ^[38]	No	Yes	No	Yes	No	No	–	No
No ^[39]	No	Yes	No	Yes	No	No	microwave	Yes
Yes^{a)}	Yes	Yes	Yes	No	Yes	Yes	terahertz	Yes

^{a)} This work

light propagation modes^[37] or making edge dispersion wind around the Brillouin zone multiple times.^[38,39] Some other explorations have focused on tuning the bandwidth and group velocity of topological kink states exhibiting slow light behavior based on the bearded interface by controlling the shape and size of air holes in the unit cells of valley photonic crystals (VPCs).^[27,28] These results suggest that topological slow light enhances light-matter interactions and is robust against extrinsic scattering loss thanks to topological protection,^[4] thereby offering a promising solution to further engineer the Q -factor of topological cavities. However, as summarized in **Table 1**, existing studies on topological cavities and slow light remain largely disconnected, with limited correlation between the two concepts. Longer delays and storage times demand resonators with higher Q -factors.^[40,41] A unified study integrating both cavity design and slow light effects in PTIs could accelerate progress in miniaturized optical technologies such as delay lines, addressable optical memories, and high-sensitivity sensors.

PTIs have been demonstrated based on the quantum Hall effect (QHE),^[11] quantum spin Hall effect (QSHE),^[42,43] and quantum valley Hall effect (QVHE).^[44–56] Among them, PTIs utilizing QHE usually require large external magnetic fields,^[11] thereby limiting their widespread applications. PTIs employing QSHE that open a topological bandgap at the Γ point, on the other hand, face the challenge of out-of-plane loss caused by leaky modes stemming from the topological edge states located above the light line.^[57] In contrast, the topological kink states of QVHE-based PTIs generated by the degeneracy-lifted K points (Dirac

points) mostly reside in the non-leaky region below the light line,^[21,27,28,54,57] providing intrinsic out-of-plane confinement due to total internal reflection. This enables the development of topological devices solely based on easily manufacturable single silicon membranes without additional substrates. Therefore, complementary metal-oxide-semiconductor (CMOS)-compatible on-chip QVHE-based PTIs can be achieved by breaking the spatial inversion symmetry of VPCs.^[52,55–60]

In this work, we investigated a series of on-chip topological valley cavities with enhanced Q -factors, including those induced by topological slow light. By interchanging the two differently sized triangular air holes in a rhombus unit cell, we constructed different interfaces from a pair of VPCs possessing broken inversion symmetry, resulting in the zigzag interfaces at the boundary with mirror symmetry, as well as the bearded interface with glide symmetry that exhibits topological slow light. In the eigenmode simulations, we found that the group index of the bearded interface topological slow light mode is several orders of magnitude larger than that of the zigzag interface. As a result, the Q -factor of the bearded interface topological cavity can theoretically exceed that of the zigzag topological cavities by several orders of magnitude in the absence of radiation loss. We also examined the influence of the cavity size on the Q -factor. A larger topological cavity provides a higher Q -factor because of the reduction in in-plane loss stemming from its larger internal bulk area, accompanied by the introduction of more resonant modes. Additionally, we employed optical excitation to tune the response of the bearded topological cavity and compared it to the zigzag topological cavity. The ex-

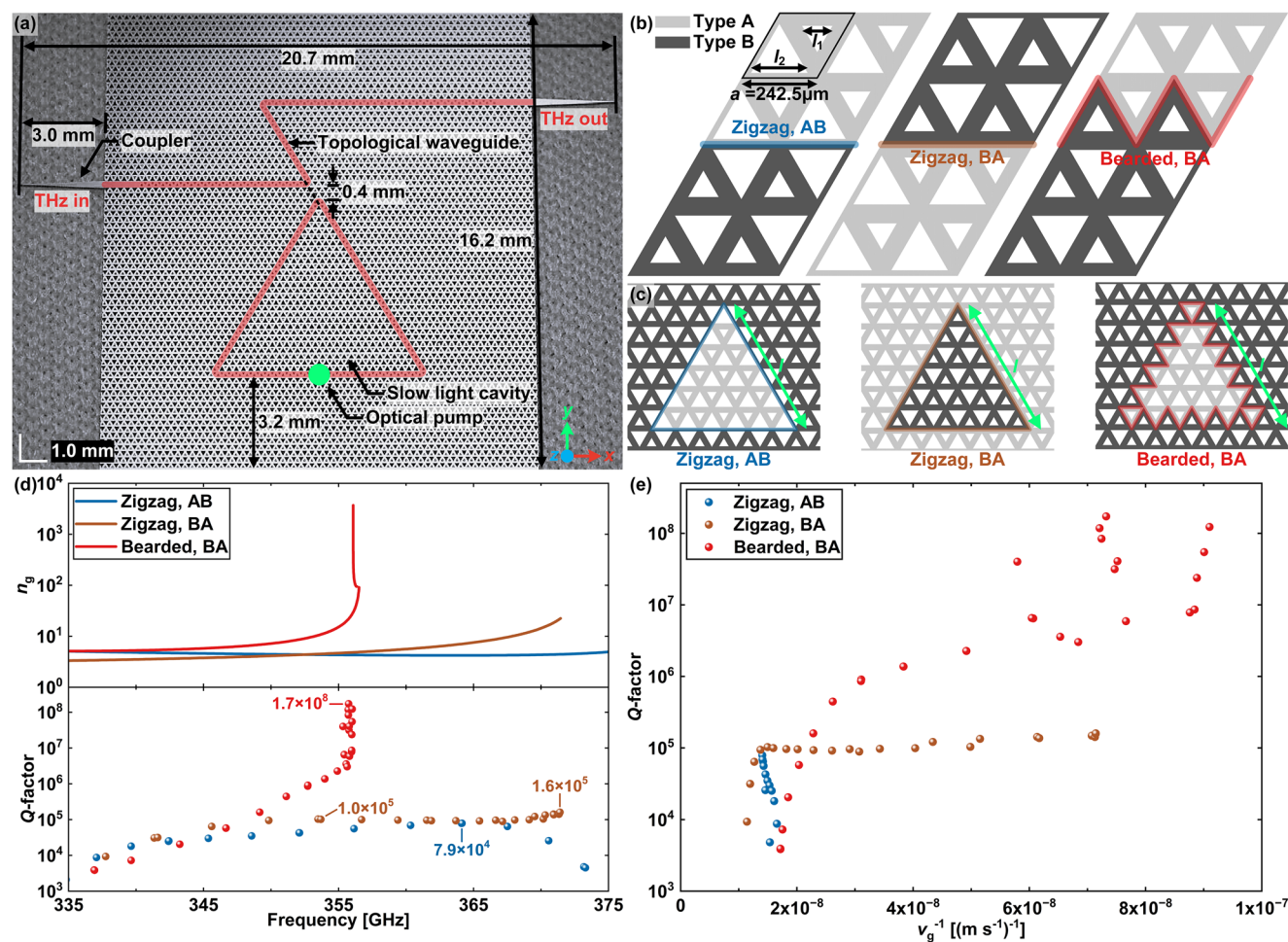


Figure 1. Design concept of the on-chip slow light cavity a) Microscope image of the fabricated valley Hall effect-based photonic topological insulator system, including a topological cavity, a topological waveguide, and two tapered couplers. b) Schematics of unit cell and interfaces of zigzag AB-type, zigzag BA-type, and bearded BA-type domain walls. The unit cell consists of two triangular air holes with side lengths $l_1 = 0.7a\sqrt{3}^{-1}$ and $l_2 = 1.3a\sqrt{3}^{-1}$, where $a = 242.5 \mu\text{m}$ is the lattice constant. c) Schematics of zigzag AB-type, zigzag BA-type, and bearded BA-type topological valley cavities. d) 2D simulation results of group indices for zigzag AB-type, zigzag BA-type, and bearded BA-type domain walls, as well as the 2D simulation results of Q -factors of eigenmodes for zigzag AB-type, zigzag BA-type, and bearded BA-type topological valley cavities with side lengths l of 30-unit cells. e) Evolution of Q -factors as functions of the inverse group velocities for zigzag AB-type, zigzag BA-type, and bearded BA-type topological valley cavities with side lengths l of 30-unit cells.

perimentally measured loaded Q -factor (Q_d) exhibits a similar trend to our theoretical predictions, which confirms the theory of slow light-induced Q -factor enhancement in topological cavities and the potential for actively controlled Q -factor modulation in slow light topological devices. Our work offers a method to facilitate the development of on-chip high- Q photonic cavities for communications, active sensors, configurable lasers, strong tunable light-matter interactions, and polaritonics.

2. Results

2.1. Design of Unit Cells, Interfaces, and Cavities for QVHE-Based PTIs

Figure 1a shows the optical image of the QVHE-based PTI sample, which consists of a topological waveguide and a topological cavity with two different VPCs. The rhombus unit cells of the two

VPCs are depicted as Type A (light gray) and Type B (dark gray) in Figure 1b, both featuring equilateral triangular air holes with side lengths $l_1 = 0.7a\sqrt{3}^{-1}$ and $l_2 = 1.3a\sqrt{3}^{-1}$, which have a common period $a = 242.5 \mu\text{m}$. When $l_1 \neq l_2$, the spatial inversion symmetry is broken, resulting in the lifting of degenerate Dirac points at K and K' valleys and the opening of a bandgap. Topologically distinct gapped bands can be characterized by different valley Chern numbers, which is a prerequisite for creating a topological kink state at the interface of the two VPCs^[4] (see detailed discussion in Section S1, Supporting Information). Here, we compare three types of interfaces and their topological interface states: zigzag AB-type, zigzag BA-type, and bearded BA-type, as highlighted by the blue, brown, and red lines at the interface between the two VPCs in Figure 1b, respectively. Figure 1c displays the details of the corresponding topological cavity structures, along with their zigzag AB-type, zigzag BA-type, and bearded BA-type interfaces

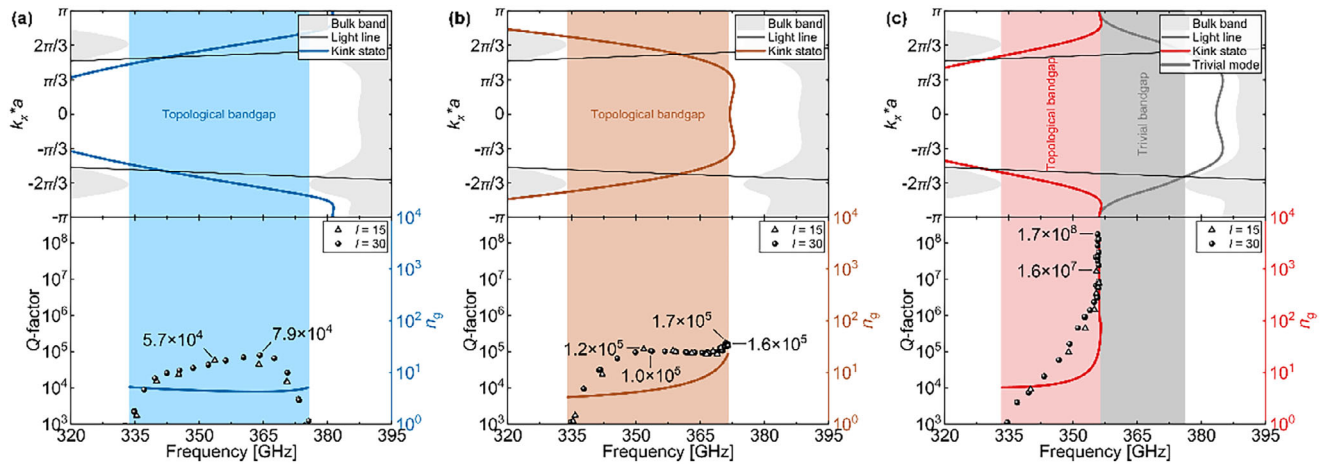


Figure 2. 2D dispersion diagrams, Q -factors of eigenmodes, and group indices of a) zigzag AB-type, b) zigzag BA-type, and c) bearded BA-type topological valley cavities. The black and gray curves represent the light line and trivial band. The blue, brown, and red curves depict the topological bands (kink states) in the top plots and group indices in the bottom plots, respectively. The light blue, light brown, and light red regions highlight the frequency range of topological bandgap, and the light gray and gray regions indicate the bulk bands and trivial bandgap, respectively. The black dots and black triangular markers in the bottom plots mark the Q -factors of eigenmodes of zigzag AB-type, zigzag BA-type, and bearded BA-type topological valley cavities with side lengths l of 30- and 15-unit cells, respectively.

marked by blue, brown, and red lines, respectively. To ensure a fair comparison of the three types of topological cavities, we set the side lengths of all cavities to $l = 30$ unit cells (see detailed discussion on bearded AB-type interface and cavity in Section S2, Supporting Information). For different interfaces, their group indices are significantly different due to their distinct types of topological states, as shown in the top plot of Figure 1d. Similarly, the visible differences are also manifested in the trends of the Q -factors of the eigenmodes of corresponding topological valley cavities enclosed by the three types of interfaces, as revealed in the bottom plot of Figure 1d.

2.2. Simulation of Different Interfacial Cavities for QVHE-Based PTIs

The 2D results of dispersion diagrams of zigzag AB-type, zigzag BA-type, and bearded BA-type interfaces illustrated in Figure 1b are shown in the top plots of Figure 2a–c, where the black curve indicates the light line, the light gray regions display the bulk bands, while the blue, brown, and red curves are their topological interface states, respectively. The topological band of the zigzag AB-type interface spans the entire bandgap, where the whole frequency range covered by the light blue area (Figure 2a), from 334 to 375 GHz, is of topological nature known as a valley kink state.^[17,51] At the lower frequencies, a bandwidth of ≈ 8 GHz is above the light line in the leaky region, which forecasts significant radiation loss of a topological system with a zigzag AB-type interface at this frequency band.^[27,56,61] For the zigzag BA-type interface (Figure 2b), its topological band fills almost the whole bulk bandgap except for a small high-frequency region above the light line. That is, the zigzag BA-type interface theoretically possesses a topological kink state within the frequency band of 334 to 372 GHz (light brown), and the radiation loss will be significantly higher at frequencies from 360 to 372 GHz. The topological kink state of the bearded BA-type interface behaves differ-

ently from the zigzag AB-type and BA-type interfaces. As shown in the top plot of Figure 2c, there are two connecting bands within the bandgap, where only the low-frequency band depicted by the red curve is topological with the entire band being situated below the light line, while the other band drawn as the dark gray curve is trivial.^[4,21] Correspondingly, the frequency bandwidth marked with the light red area (from 334 to 356 GHz) supports a robust topological kink state.

The group index $n_g = c/v_g$ can be obtained from the slope of the topological band v_g ,^[19,20,28] where c is the speed of light in vacuum. The 2D simulated results of group indices of zigzag AB-type, zigzag BA-type, and bearded BA-type domain walls are represented by blue, brown, and red curves in the bottom plots of Figure 2a–c, respectively. For the zigzag AB-type interface, its group index varies only slightly throughout the entire topological bandgap. In contrast, for the zigzag BA-type interface, the group index increases exponentially as the frequency increases across the topological bandgap, up to an order of magnitude. Similarly, for the bearded BA-type interface, the value of its group index increases by several orders of magnitude as the frequency rises within its topological bandgap, especially toward the high-frequency edge of the topological band. Therefore, both the zigzag BA-type and bearded BA-type domain walls exhibit the property of slow light, with the bearded interface demonstrating a larger group index, while the zigzag AB-type domain wall possesses no slow light characteristic. The group indices of the three types of interfaces are summarized in the top plot of Figure 1d to clearly demonstrate the differences in their slow light behavior.

To investigate the influence of different interfaces on the topological cavities, as portrayed in Figure 1c, we designed three types of topological cavities featuring zigzag AB-type, zigzag BA-type, and bearded BA-type boundaries, where the side lengths of all topological cavities are set to $l = 30$ unit cells. We employed 2D simulations to exclude radiation loss, and the eigenmode results are depicted with black dots at the bottom of Figure 2a–c. Among them, the Q -factors of the eigenmodes of the zigzag AB-type

topological cavity initially increase and then decrease with higher frequency within the topological bandgap, as shown in the bottom plot of Figure 2a. The maximum Q -factor appears within the topological bandgap, as the cavity modes situated near the bulk bands suffer from high in-plane loss due to scattering into the bulk. For the zigzag BA-type topological cavity, the Q -factors of its eigenmodes displayed in the bottom plot of Figure 2b initially increase, subsequently decrease and finally increase again with the rise in frequency throughout the topological bandgap, and the two local maximum Q -factors are located at the center and the high-frequency edge of the topological bandgap, respectively. The occurrence of the first local maximum Q -factor here is primarily attributed to low in-plane loss, similar to the case of the zigzag AB-type topological cavity. The trend of the Q -factors within the high-frequency range of the topological bandgap exhibits a strong correlation with the trend of the group index, marked by the second local maximum Q -factor (which is also the highest Q -factor) and the slow light extremum that both appear at the high-frequency edge of the topological bandgap, indicating that the second local maximum Q -factor arises from the enhanced light-matter interactions induced by slow light. In addition, the values of the two local maximum Q -factors visually demonstrate that the contribution of slow light far exceeds that of low in-plane loss for enhancing the Q -factors of zigzag BA-type topological cavities. This conclusion is also supported by the bearded BA-type topological cavity. As clearly shown in the bottom plot of Figure 2c, the narrow band characterized by a sharp rise in group index is also the region where the Q -factors increase dramatically for the bearded BA-type topological cavity. This narrow band, in comparison to the center of the topological bandgap in the zigzag BA-type topological cavity, is located not only in the middle of two bulk bands but also at the high-frequency edge of the topological bandgap, implying that the Q -factors in this region should be jointly enhanced by the low in-plane loss and slow light nature.

Quantitatively, as illustrated in the bottom plot of Figure 1d, the maximum Q -factor arising from low in-plane loss within the topological bandgap of the zigzag BA-type topological cavity ($Q_{2D} = 1.0 \times 10^5$) shows a 26.6% increase compared to that of the zigzag AB-type topological cavity ($Q_{2D} = 7.9 \times 10^4$). The increase in Q -factor is due to the slow light effect in the zigzag BA-type cavity. Moreover, this enhancement of Q -factor surpasses twofold when considering the maximum Q -factor at the high-frequency edge of the topological bandgap of the zigzag BA-type cavity ($Q_{2D} = 1.6 \times 10^5$), which is directly supported by the slow light phenomena. The bearded BA-type topological cavity demonstrates superior performance with a maximum Q -factor of $Q_{2D} = 1.7 \times 10^8$, combining slow light propagation with low in-plane losses. This represents 1699 times higher than the maximum Q -factor derived from low in-plane loss in the zigzag BA-type topological cavity. Simultaneously, it shows a 1062 times improvement over the latter's maximum Q -factor corresponding to slow light and low in-plane loss (high-frequency edge of the topological bandgap). In comparison to the maximum Q -factor of the zigzag AB-type topological cavity, this represents an enhancement by a factor of 2151. As a result, the zigzag BA-type topological cavity offers a better design for achieving higher Q -factors compared to the zigzag AB-type topological cavity, while the bearded BA-type topological cavity emerges as the

best topological edge state platform due to the combined effect of slow light and low in-plane loss with slow light playing a pivotal role. While the zigzag BA-type topological cavity does show Q -factor enhancement from slow light effects at the high-frequency bandgap edge, its actual performance in this region may deviate from eigenmode predictions. These frequencies fall within the leaky regime, potentially causing resonances to disappear.

In order to ascertain the relationship between the side length of a topological cavity and its Q -factor, we also performed simulations of the eigenmodes of the zigzag AB-type, zigzag BA-type, and bearded BA-type topological cavities with $l = 15$ unit cells, as indicated by the black triangular markers in the bottom plot of Figure 2a–c. For the zigzag AB-type and BA-type cases (Figure 2a,b), halving the side length of the cavity leads to a significant reduction in the number of eigenmodes, while the corresponding Q -factors exhibit only limited variation. For the bearded BA-type cavity with $l = 15$ unit cells (Figure 2c), its Q -factors of the eigenmodes follow a similar increasing trend with frequency as compared to those for $l = 30$ unit cells, albeit with fewer eigenmodes for the shorter side length. Ultimately, the maximum Q -factor of the cavity with $l = 15$ unit cells stands on the order of 10^7 , which is one order of magnitude lower than that of the cavity with $l = 30$ unit cells. Here, the lower Q -factor can be attributed to the increased in-plane scattering resulting from its smaller bulk area inside the cavity. Larger topological cavities, thus, provide higher Q -factors, at the cost of larger device sizes with more resonant modes within the topological bandgap due to the reduced free spectral range. Both of these factors are disadvantageous for practical applications. In other words, a trade-off between the Q -factor and the size is crucial when developing a topological cavity-based device with a specific function.

For a clearer visualization of the relationship between the Q -factor and slow light, the Q -factor versus the inverse group velocity for the three types of topological cavities with side length l of 30-unit cells is plotted in Figure 1e. Evidently, the Q -factors of the zigzag AB-type topological cavity (blue dots) are confined to a small range of low vg^{-1} values due to its lack of slow light characteristics. By contrast, the Q -factors of the zigzag BA-type topological cavity (brown dots) cover a broad range of the horizontal axis. Beyond a small range of low vg^{-1} values, the Q -factors remain nearly constant with increasing vg^{-1} , reflecting the limited enhancement effect of weak slow light properties on the Q -factors of the cavity. For the bearded BA-type topological cavity, the Q -factors are distributed over a wider range of the horizontal axis compared to those of the zigzag BA-type topological cavity and exhibit an overall upward trend as vg^{-1} increases. These results align with the fact that the bearded BA-type topological cavity possesses better slow light performance, highlighting the direct role of slow light in enhancing Q -factors in slow light topological cavities.

2.3. Experiment with Cavities for QVHE-Based PTIs

To verify our hypothesis that slow light and low in-plane loss enhance the Q -factor and slow light plays a greater role, we designed several topological valley cavities on a silicon (Si) chip and measured their loaded Q -factor Q_{ld} . The sample containing a bearded BA-type topological cavity is depicted in Figure 1a,

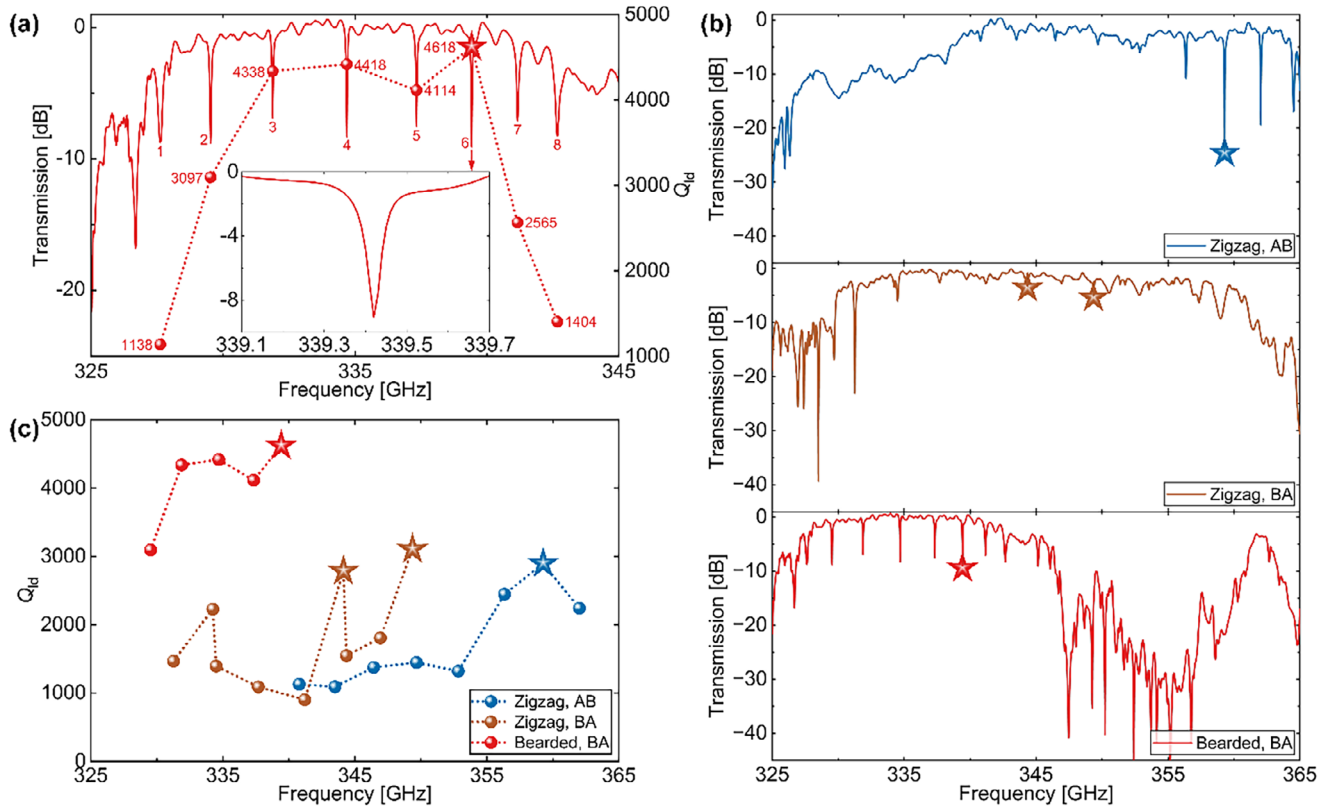


Figure 3. Measured transmission of the on-chip cavities. a) Experimental transmission spectrum and corresponding loaded Q -factors for bearded BA-type topological cavity. b) Experimental transmission spectra (see details in Section S4, Supporting Information) and c) calculated loaded Q -factors for zigzag AB-type, zigzag BA-type, and bearded BA-type topological valley cavities. The pentagrams indicate the maximum Q -factors and their corresponding frequencies.

which also incorporates a bearded BA-type topological waveguide designed to excite the cavity and two tapered couplers are set at both ends of the topological waveguide to optimize the transmission of terahertz waves. The side length of the cavity is $l = 30$ unit cells, and its distance from the Z-shaped topological waveguide is $d = 2$ unit cells (≈ 0.4 mm) along the extended line of the waveguide at its sharp bend, with a separation of 15-unit cells (≈ 3.2 mm) from the lower edge of the sample. The dimensions of the sample are 20.7 mm in width and 16.2 mm in length, incorporating two tapered couplers each with a length of 3 mm. To make a fair comparison, we also designed samples of zigzag AB-type and zigzag BA-type topological cavities excited by Z-shaped zigzag AB-type and BA-type topological waveguides with $l = 30$ unit cells using a similar scheme, respectively (see Section S3, Supporting Information). All samples are fabricated on a silicon wafer with a thickness of $t = 220$ μm and a resistivity of $>10\,000$ $\Omega\cdot\text{cm}$. Figure 3a shows the experimental transmission and the extracted Q_d of corresponding resonances for the bearded BA-type topological waveguide-cavity system. The high transmission exhibits a topological kink state in the range from 327 GHz to 345 GHz, although there exists a slight frequency shift and bandwidth difference as compared to the topological bandgap of 334 GHz to 356 GHz in Figure 2c. The observed discrepancies are likely owing to fabrication errors in the photolithography process and variations in the wafer thickness. Despite this, the measured transmission is above -1 dB from

329 GHz to 341 GHz, and the Q_d of resonances within this range exhibits an increasing trend with frequency, aligning with the trends of Q -factors and group index presented in the bottom plot of Figure 2c. For lower and higher frequencies near the edge of the transmission bandwidth, the Q_d of resonances would no longer be accurate due to their low transmission.

We also conducted experiments on the zigzag AB-type and zigzag BA-type topological waveguide-cavity systems, and the transmission spectra as well as Q_d of their resonances along with those of the bearded BA-type waveguide-cavity system are plotted together in Figure 3b,c, respectively. Wherein Q_d of those resonances falling within the low transmission ranges remain ignored because of their inaccuracy. It is evident that the Q_d for the zigzag AB-type topological cavity (blue dots) shows a trend of gradual increase followed by a rapid decline as frequency increases, which is in agreement with the trend of Q -factors depicted in the bottom plot of Figure 2a. For the zigzag BA-type topological cavity, a local maximum value ($Q_d = 2,786$) is observed at 344 GHz, and the Q_d continue to increase at high-frequency band. Moreover, the Q_d of zigzag BA-type topological cavity are generally larger than that of zigzag AB-type topological cavities within the same frequency range and the maximum Q_d of bearded BA-type topological cavity significantly surpasses both the values of zigzag AB-type and BA-type cavities, reaching a value of 4618. Despite the Q -factors achieved from the loaded system being several orders of magnitude lower than those ex-

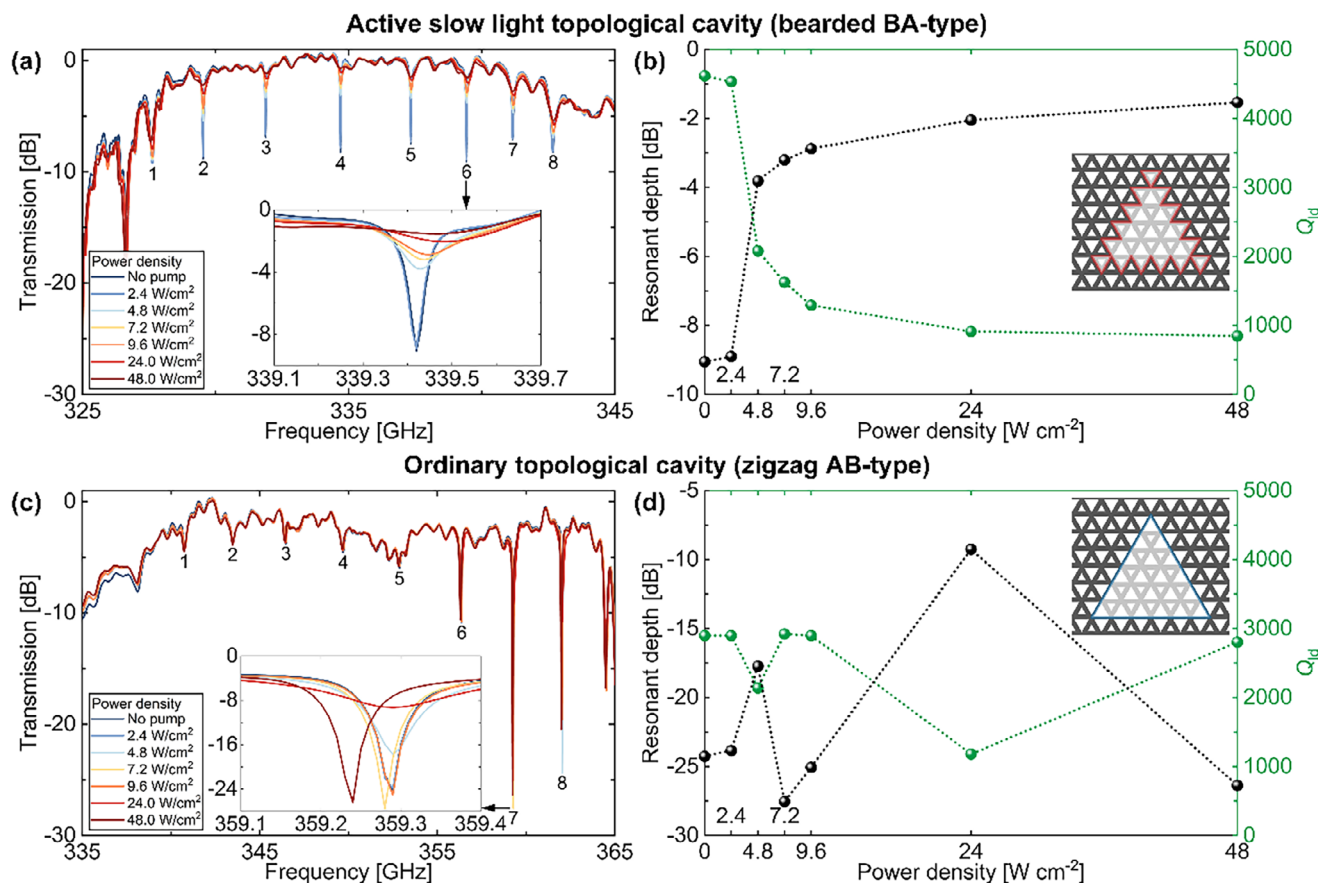


Figure 4. Active optical control of the on-chip cavities. a) Experimental transmission spectra, b) resonant depths and loaded Q -factors of the 6th order resonance for bearded BA-type topological waveguide-cavity system with different pump power densities. c) Experimental transmission spectra, d) resonant depths, and loaded Q -factors of the 7th order resonance for zigzag AB-type topological waveguide-cavity system with different pump power densities. The illustrations in (b) and (d) are schematics of bearded BA-type and zigzag AB-type topological valley cavities, respectively.

tracted from the eigenmodes (see discrepancy analysis in Section S5, Supporting Information), these experimental observations of Q -factors agree well with the trends of the theoretical predictions, where higher Q -factors are obtained for the zigzag BA-type and bearded BA-type topological cavities, especially toward higher frequencies of the slow light regime.

Active Q -factor control is extremely important for many important applications including communications, lasers, and nonlinear optics. In view of this, we further examined the active device potential of different types of topological valley cavities. Taking the zigzag AB-type and bearded BA-type topological cavities as examples, a photoexcitation with pump power density ranging from 0 W cm⁻² to 48.0 W cm⁻² was applied at the midpoint of the bottom edge of the cavities (green circle position in Figure 1a), and the corresponding results are displayed in Figure 4. For the bearded BA-type topological waveguide-cavity system, as shown in Figure 4a, all resonances weaken monotonically in magnitude as the pump power density increases. This change is caused by the photoexcitation of Si above the bandgap to generate free carriers, which absorb terahertz waves and thereby alter the inherent loss rate of the cavity. Consequently, the larger the power density of photoexcitation, the higher the loss of terahertz waves propagating along the domain wall of the topological cav-

ity when passing through the irradiated region, causing the resonance to weaken until it disappears. As an example, we study the 6th order resonance clearly as illustrated in the zoom-in inset of Figure 4a. To further quantify this trend, we plotted the resonant depth and corresponding Q_d of 6th order resonance under different pump power densities in Figure 4b. The resonant depth (black dots) rapidly decays as the pump power density increases from 0 W cm⁻² (without photoexcitation) to 4.8 W cm⁻², which is attributed to the generation of free carriers. This weakening trend gradually slows down as the pump power density continues to rise, particularly when the power density ranges from 4.8 W cm⁻² to 9.6 W cm⁻² and from 9.6 W cm⁻² to 48.0 W cm⁻², where the resonant depth exhibits nearly linear changes. Following this trend, the Q_d (green dots) also becomes lower with increasing pump power density, and it drops almost linearly in both cases from 4.8 W cm⁻² to 9.6 W cm⁻², and from 9.6 W cm⁻² to 48.0 W cm⁻². This occurs because the additional free carriers excited by the pump light in the selected single-crystal silicon become increasingly scarce within a finite region. The experimental results show modulations in resonant depth ranging from -9.1 dB to -1.5 dB and Q -factor from 4618 to 848 using an optical pump power density of as low as 48 W cm⁻². This demonstrates the active tunability of ≈ 6.3 W cm⁻² per dB in resonant depth for the

slow light topological cavity. All different order resonance of the bearded BA-type topological cavity shows a similar trend.

Unlike the bearded BA-type topological cavity, the response of the zigzag AB-type topological cavity to photoexcitation is significantly different. Figure 4c shows the transmission of the zigzag AB-type topological waveguide-cavity system with and without the optical pump. Compared with the bearded BA-type waveguide-cavity system, the transmission of the zigzag AB-type waveguide-cavity system remains low from 335 GHz to 340 GHz. As mentioned previously, this phenomenon primarily arises from the fact that the zigzag AB-type topological band lies above the light line here, thus resulting in a high loss. In addition, the responses of the 1st to 5th order resonances are weak, which can be explained due to the imperfect coupling caused by an incomplete matching distance d between the cavity and waveguide as well as fabrication errors. Besides, to gain a deeper understanding of the optical response of the zigzag AB-type topological cavity, its transmission, resonant depth, and Q_d of the 7th order resonance for different pump power densities are shown in Figure 4c,d, since the 7th order resonance exhibits the lowest transmission depth. When the pump power density increases from 0 W cm⁻² (without photoexcitation) to 48.0 W cm⁻², as displayed in the inset of Figure 4c, the resonant depth of the 7th order resonance oscillates repeatedly and even shifts in frequency. This phenomenon occurs because photoexcitation causes the zigzag AB-type topological cavity to transition between multiple coupling states: undercoupling, critical coupling, and overcoupling, due to an incomplete matching distance d . Additionally, the thermally induced changes in the cavity are more pronounced in the zigzag AB-type structure compared to the bearded BA-type topological cavity.^[17] Figure 4d shows more clearly the oscillation in both the resonant depth and Q_d of the 7th order resonance across the entire pump power density range from 0 W cm⁻² to 48.0 W cm⁻². When comparing the optical responses of bearded BA-type and zigzag AB-type topological cavities (both with a 2-unit cell coupling distance) to their respective topological waveguides, it becomes evident that the bearded BA-type cavity exhibits a smoother, nearly linear response to varying photoexcitation intensities. On this basis, we can further predict that the bearded topological cavity possesses better linear responsivity than the zigzag topological cavity in the face of changes in external excitation or surrounding environment. That is, the high- Q slow light topological cavity demonstrates a greater degree of control for both active and passive manipulation.

3. Discussion

In conclusion, we have conducted a comprehensive investigation of zigzag AB-type, zigzag BA-type, and bearded BA-type interfaces in valley Hall topological insulators and their corresponding topological valley cavities, systematically elucidating the critical relationship between slow light effects and cavity Q -factors. Our analysis began with detailed dispersion characterization of these interfaces, revealing that while both zigzag BA-type and bearded BA-type domain walls exhibit slow light properties (with particularly high group indices in the bearded case), the zigzag AB-type interface shows no slow light behavior. Through combined 2D simulations and experimental validation, we demonstrate that optimal Q -factor enhancement arises from the synergistic com-

bination of slow light propagation and low in-plane losses, with slow light effects playing the dominant role. The bearded BA-type cavity emerges as the superior configuration, achieving a maximum loaded Q -factor of 4618, significantly outperforming the zigzag-type cavity's 3106, by simultaneously leveraging both enhancement mechanisms at the same operational frequency. Through optical active control experiments, we further demonstrate the superior modulation capabilities of the bearded topological valley cavity, exhibiting enhanced sensitivity to external stimuli with a tunable response of 6.3 W cm⁻² per dB in resonant depth. Furthermore, our results highlight an important engineering consideration for application-specific topological cavity designs: the critical trade-off between achieving maximum quality factor and maintaining compact device dimensions must be carefully optimized based on target performance metrics. The bearded cavity configuration proves advantageous in this regard by simultaneously offering both high Q -factor maintenance and strong field confinement. These on-chip topological valley cavities, leveraging slow light effects to achieve high Q -factors, open new possibilities for advanced integrated photonics, from spectral-efficient multiplexers and sensitive detectors to tunable coherent sources and miniaturized LiDAR platforms.

4. Experimental Section

Sample Fabrication: The VPC chips were fabricated on a 4-inch diameter 220 μ m thick high-resistivity silicon (Si) wafer with a resistivity of >10 k Ω -cm. The Si wafer was carefully cleaned, followed by the deposition of a 2 μ m thick silicon-dioxide (SiO₂) layer. Photolithography was used to define the triangular etch holes and device separation trenches. SiO₂ was selectively etched away, exposing the Si surface. To achieve the uniform side wall of holes and avoid over-etching, deep reactive ion etching of Si up to a depth of 200 μ m was carried out. The photoresist layer and SiO₂ layers were then completely removed.

Numerical Simulation: Frequency domain simulation was used to process the 2D calculation of dispersion and eigenmode. For dispersion simulation, Floquet periodic and scattering boundary conditions were imposed in the direction of the topological interface extension and its perpendicular direction, respectively. For eigenmode simulation, the outer VPC region extends 15-unit cells outward from the inner VPC region, and scattering boundary conditions were applied to its three bounding edges. The group index was calculated by extracting the topological band from dispersion data.

Experimental Setup: A signal generator and a spectrum analyzer with frequency extension modules were used to characterize the topological waveguide-cavity samples. The microwave frequency through the signal generator was upconverted to the desired frequency band (0.33 THz to 0.5 THz) by a frequency-multiplier-module with a WR-2.2 hollow waveguide for the THz wave output port. The tapered coupler of the sample was used to couple the THz waves from the WR-2.2 waveguide. THz transmission through the sample was again outcoupled to signal-analyzer-module through the WR-2.2 waveguide to down convert the incoming signal for spectrum analysis. The signal analyzer-module upconverted the local oscillator (LO) signal generated from the spectrum analyzer and mixed it with the incoming THz signal to generate an intermediate frequency (IF) signal. To obtain the normalized spectra, the WR-2.2 waveguides of both modules were first directly connected (without samples) to measure the signal strength in the desired frequency range.

For photoexcitation, a continuous laser of wavelength 532 nm ($E = 2.36$ eV) was used, which has a photon energy higher than the bandgap of silicon (1.1 eV). The laser beam was further focused using a convex lens onto the topological waveguide-cavity chip. The beam size was kept small to illuminate only the domain wall of the cavity (≈ 0.73 mm of the beam

spot diameter) with the pump density ranging from 0 W cm⁻² to 48 W cm⁻², as shown in Figure 1a.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

X.Z., Y.J.T. and R.S. conceived the idea. X.Z., Y.J.T., and W.W. performed the schematic design. X.Z. executed the numerical simulations. X.Z., Y.J.T., W.W., J.J.Y., L.N.Z. and R.S. implemented the theoretical analysis. X.Z. and Y.J.T. fabricated the silicon devices. Y.J.T., W.W., and K.J.C. carried out the experiments. X.Z. completed the numerical calculation with Z.L.S. and S.X.W., X.Z., Y.J.T. and J.J.Y. accomplished the data analysis. X.Z. and R.S. wrote the manuscript with inputs from Y.J.T., W.W., J.J.Y., and L.N.Z. R.S. led the overall project.

Data Availability Statement

The authors declare that the data supporting the findings of this study are available within the paper and its supplementary information files.

Keywords

high-Q cavity, interfacial waveguide, on-chip device, slow light topological cavity, terahertz silicon photonics, terahertz topological photonic integrated circuit

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